MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

DEPARTMENT OF CIVIL ENGINEERING			
END SEMESTER EXAMINATION		**NOVEMBER, 2024**	
Level	**B.Tech**	**Curriculum**	**2022**
Program Name	**B.Tech**	**Course Type**	**PE**
Course Name	**Elements of Earthquake Engineering**	**Course Code**	**CIE 4078**
Semester	**VII**	**Max Marks**	**50**
Date	**22/11/2024**	**Duration**	**180 minutes**

Instructions:

- **Answer ALL the questions.**
- **Data not provided may be suitably assumed**
- **Usage of IS 1893:2016**

Q. No.	Question	Marks	NBA		BL
			CO	PO	
1A.	Explain P-Wave shadow zone.	**2**	**CO1**	1,2,3,12, PSO2	2
1B.	In a SDOF system, the spring deflects 75 mm when the weight of 1000 N is supported on it. And a damper of resistance 600 N-sec/m is provided for the system. Determine, a) Natural frequency, b) damped frequency, c) Damping ratio, d) logarithmic decrement.	**3**	**CO2**	1,2,3,12, PSO2	3
1C.	Define dynamic magnification factor and derive the expression of maximum dynamic magnification factor.	**5**	**CO3**	1,2,3,12, PSO2	2
2A.	The static deflection of an electric generator of mass 550 kg supported by a system of four parallel springs is found to be 60 mm. If the electric generator has a rotating unbalance of 0.6 kg-m, determine the amplitude of vibration at 2000 rpm.	**5**	**CO3**	1,2,3,12, PSO2	3
2B.	A frame shown in **Fig.1** supports a rotating machine, which exerts a horizontal force at the girder level is F(t) = 350 N sin 15 Hz t. Assuming 5% of critical damping, determine steady-state amplitude of vibration and the maximum dynamic bending stress in the column Take E = 200 kN/mm^2, I = $4x10^6$ mm^4 and Z = 2000 mm^3 for each leg.	**5**	**CO3**	1,2,3,12, PSO2	3
3A.	A frame shown in **Fig. 2**, supports a motor of mass 120 kg. The mass of floor on which the motor is mounted is 5500kg. The motor has an unbalanced force of 12 kg-m and operates at 1000 rpm. Determine the maximum amplitude of displacement and maximum bending moments in each column. Assume 15% damping and take EI=4000 $kN-m^2$.	**5**	**CO3**	1,2,3,12, PSO2	3

3B.	An electric motor weighing 2000 N and operating at 450 rpm is mounted on 4 parallel springs of stiffness 4000 N/m each. Determine the maximum permissible unbalance in order to limit the steady state amplitude of 4mm and assume damping as 10% of critical damping.	**3**	**CO3**	1,2,3,12, PSO2	3
3C.	Explain frequency ratio.	**2**	**CO3**	1,2,3,12, PSO2	2
4A.	Determine base shear and its distribution for all the floors using equivalent static load method for the **Fig.3**. The structure is a residential apartment with more than 200 occupants, built by ordinary RC frames on medium strength soil in Bangalore. Wall panels are not present in the building. The beam dimension is 230×400mm, columns are 230×500mm and slab is 150 mm thick. Live load on slab is 2.5 kN/m^2.	**8**	**CO4**	1,2,3,12, PSO2	3
4B.	Draw strorey shear and base shear diagram for the 4A. question.	**2**	**CO4**	1,2,3,12, PSO2	2
5A.	Derive the amplitude of SDOF system subjected to force due to rotational unbalance.	**4**	**CO3**	1,2,3,12, PSO2	2
5B.	Explain the importance of longitudinal bars lapping location in beams with neat sketch.	**4**	**CO5**	1,2,3,12, PSO2	2
5C.	Explain slab strengthening.	**2**	**CO5**	1,2,3,12, PSO2	2

MARK DISTRIBUTION (CO-wise & BTL-wise)

CO / CLO	Total Marks
1	**2**
2	**3**
3	**29**
4	**10**
5	**6**

BL	Total Marks
2	**21**
3	**29**
4	
5	

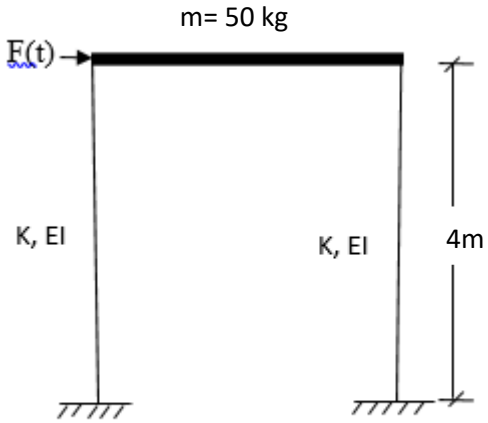


Fig. 1

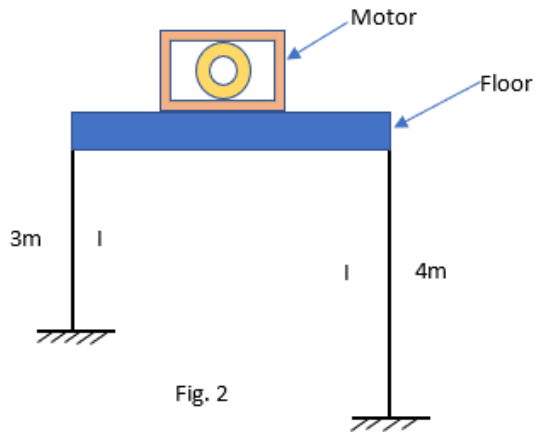


Fig. 2

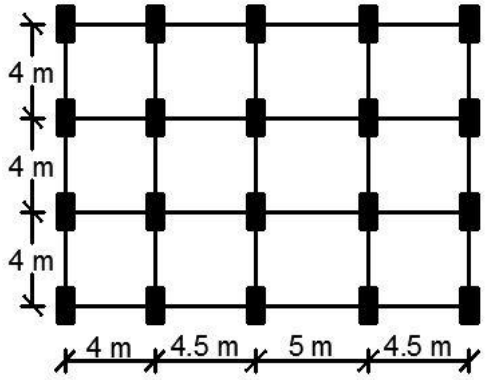


Fig. 3

